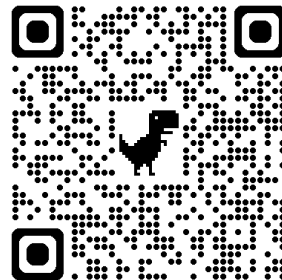


# Coarse-to-Fine Breast Lesion Segmentation in Ultrasound Images: A Detection-Guided Two-Stage Framework

Aamal Alghamdi<sup>1,2</sup> and Mohamed Elawady<sup>1</sup>

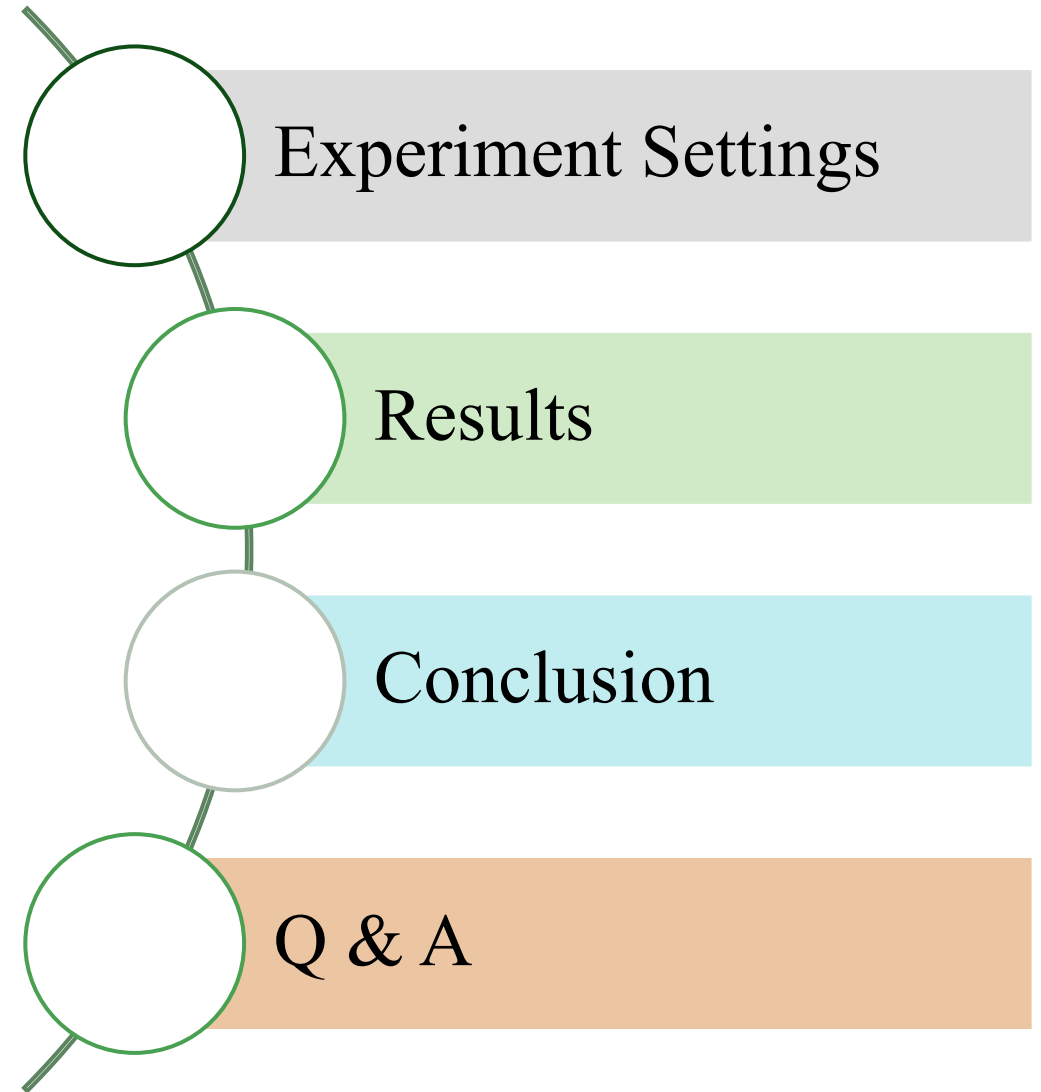
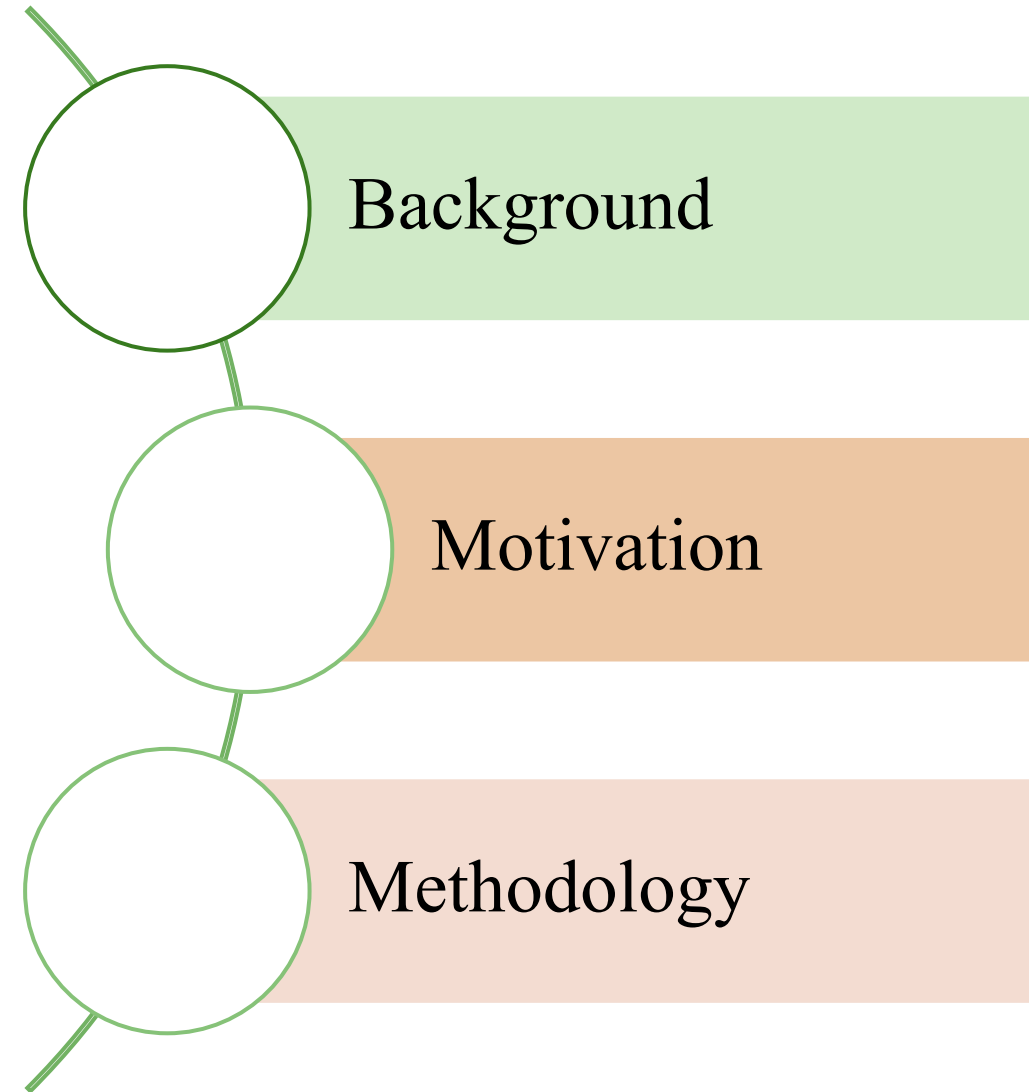
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<sup>2</sup>Department of Computer Science, College of Computers and Information Technology, Taif University, Taif, Saudi Arabia



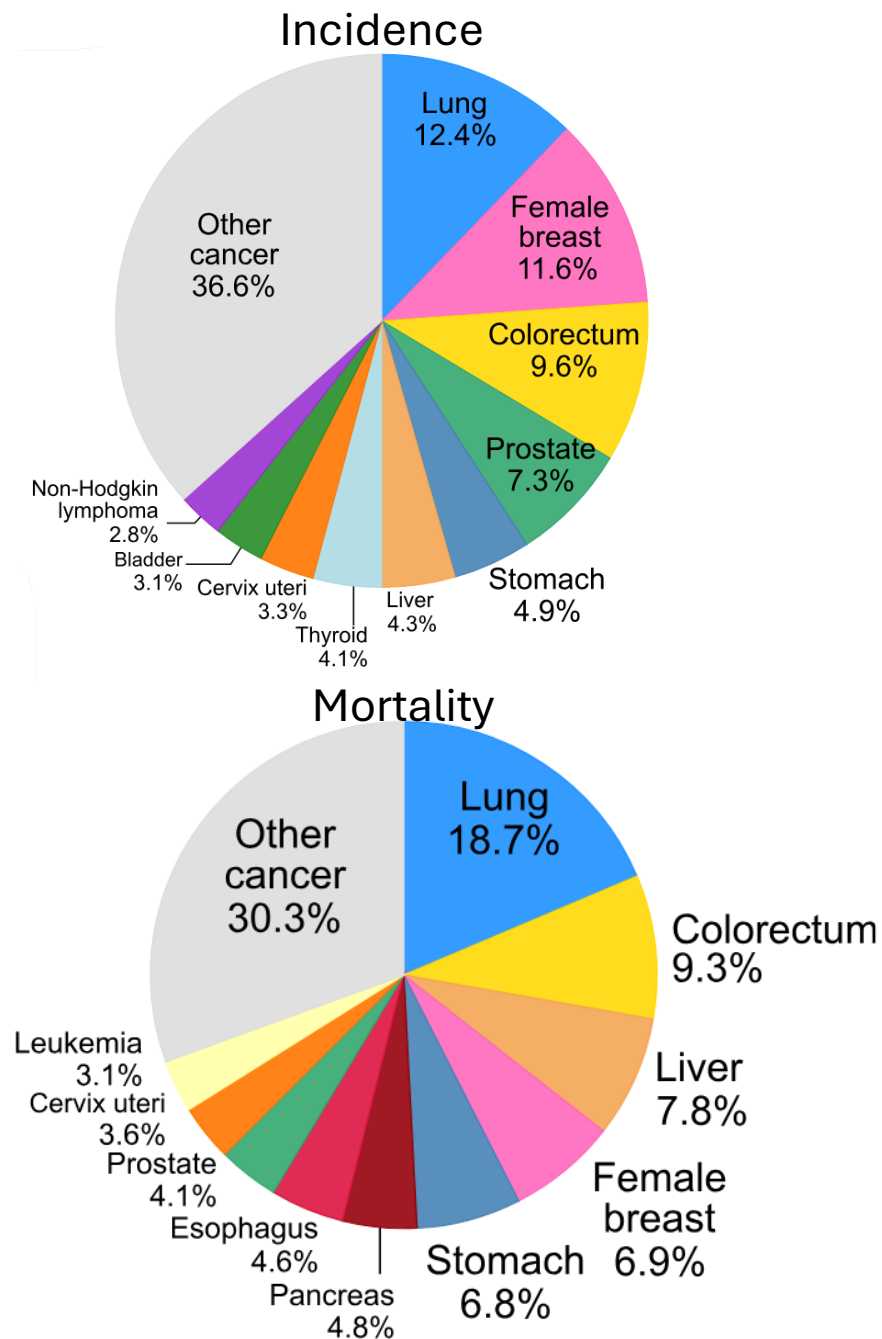
<https://github.com/mawady/miua26>

# Outline



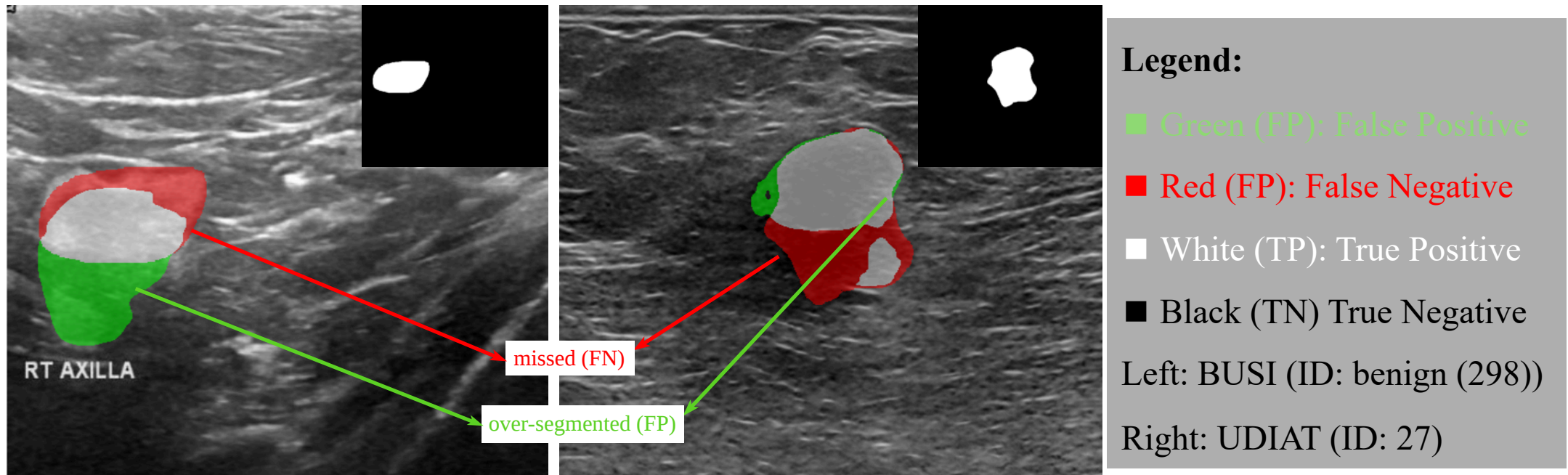
# Background

- Breast Cancer (BC) is the **2nd most common cancer** worldwide.
  - 2.3M new cases in 2022
- **leading cause of cancer death among women.**
  - 666,000 deaths
- **Early detection, accurate identification, and access to treatment** are key to recovery and reducing mortality.
- Late-stage detection is common due to BC's subtle early presentation, **reducing survival potential.**
- Diagnosis relies on **medical imaging** to help radiologists analyse the human body, but manual segmentation is **time-consuming, error-prone,** and risks misdiagnosis.



# Motivation

- Artificial Intelligent & Computer Aided Diagnosis systems enhance diagnostic accuracy.
- Existing segmentation models process the entire BUS image.
- Lesions occupy only a small fraction of the image → Severe class imbalance.
- Models learn background features more than lesion characteristics → Results in poor segmentation of lesion boundaries.
- Need for models that focus specifically on the lesion region.



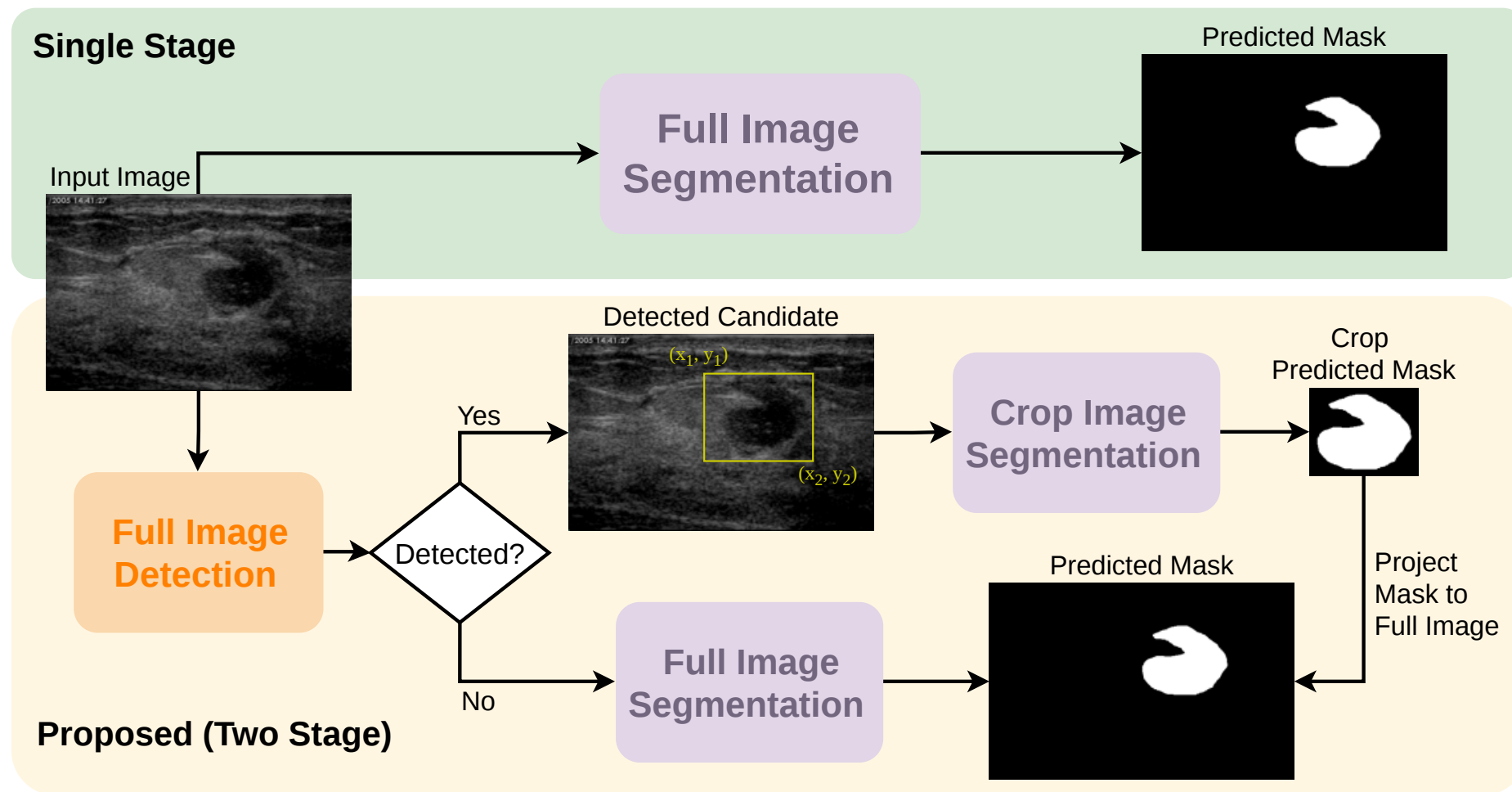
UDIAT: Yap et al., IEEE J. Biomed. Health Inform., 2017.

BUSI: Al-Dhabyani et al., Data in Brief, 2020.



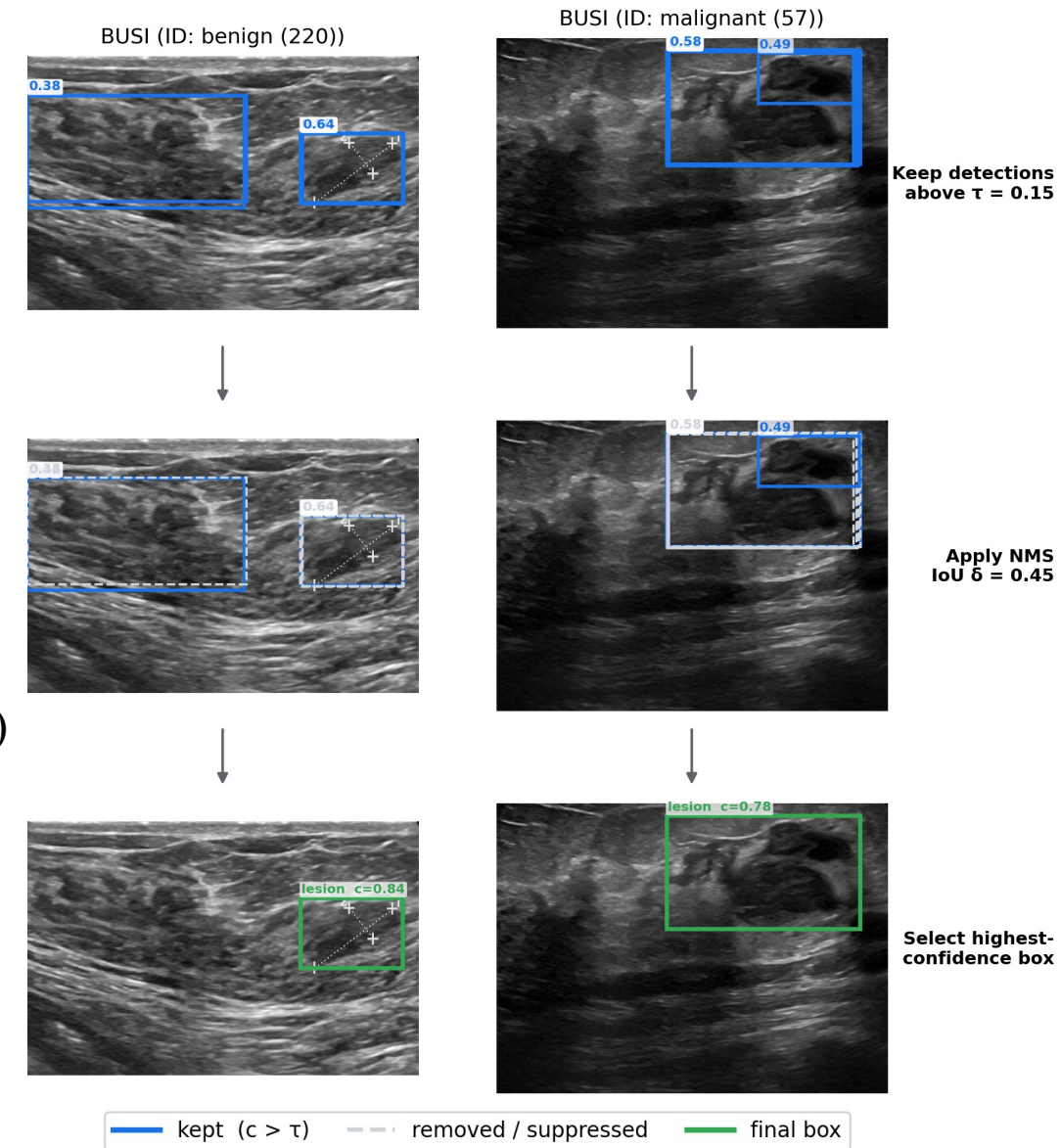
# Methodology

- **Detection-guided segmentation:** YOLO crops focus a segmentation network, reducing false positives.
- **Comprehensive evaluation:** validated on UDIAT, BUSI, and BUSBRA with detection + segmentation metrics.



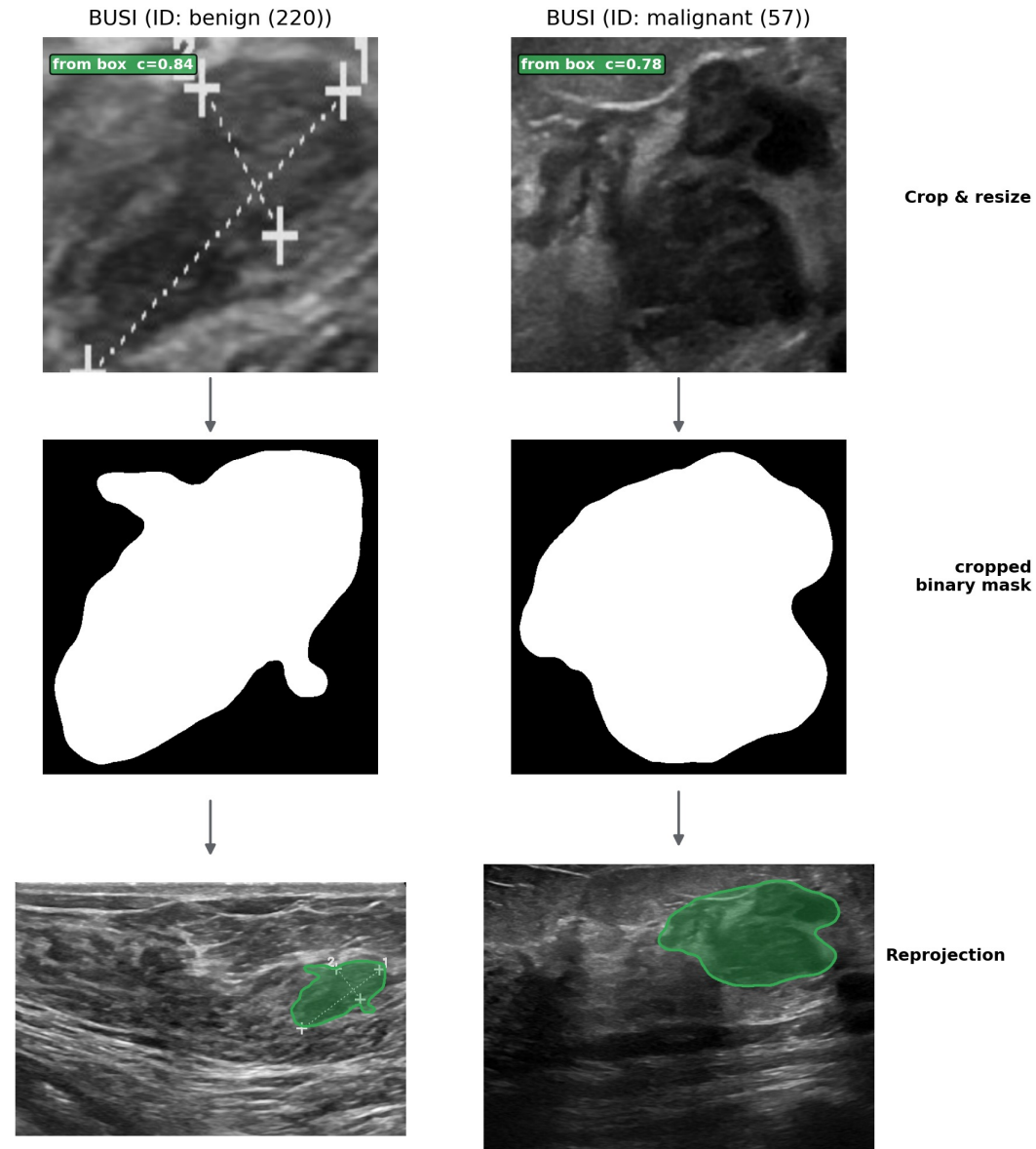
# Methodology: Stage 1 - Lesion Localisation

- **Task:**
  - An object detector localises lesions in the full image.
  - Predicting a bounding box with confidence  $c$ .
- **Filtering:**
  - Keep detections above threshold  $\tau = 0.15$ .
  - Apply Non-Maximum Suppression ( $\text{IoU } \delta = 0.45$ ).
  - Select the highest-confidence box.
- **Models:** SOTA detectors widely used in medical imaging.
  - YOLOv3-tiny (8.9M), YOLOv8s (11.2M), YOLOv12s (9.3M)
- **Rationale:**
  - Tiny/small variants: strong speed–accuracy trade-off.
  - Similar parameter counts ensure a fair comparison.

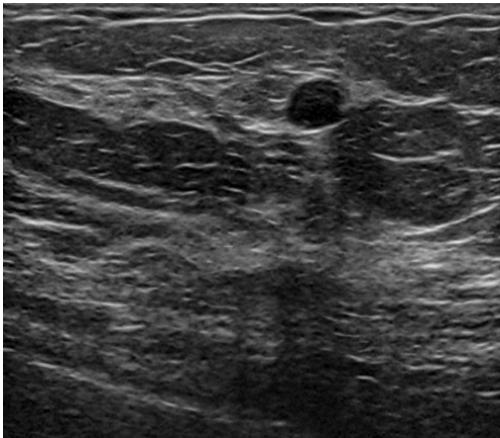


# Methodology: Stage 2 - Lesion Delineation

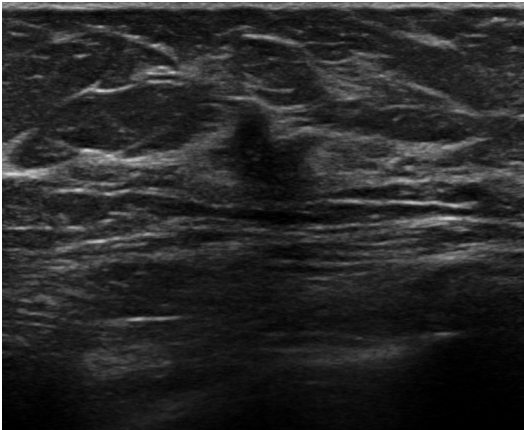
- **Crop:**
  - Extract a tight crop from the detected box and resize it.
  - Pass it to a segmentation network trained on cropped lesions.
- **Mask generation:**
  - The final output of the segmentation method is a cropped binary mask.
- **Reprojection and Fallback:**
  - Resize cropped image back.
  - Place it onto a zero canvas to form the full-image prediction.
  - If no confident detection, a full-image segmentation model runs on the whole input.
- **Models:** SOTA segmentation widely used in medical imaging
  - U-Net, UNet++, DeepLabV3+.



# Experimental Settings: Datasets



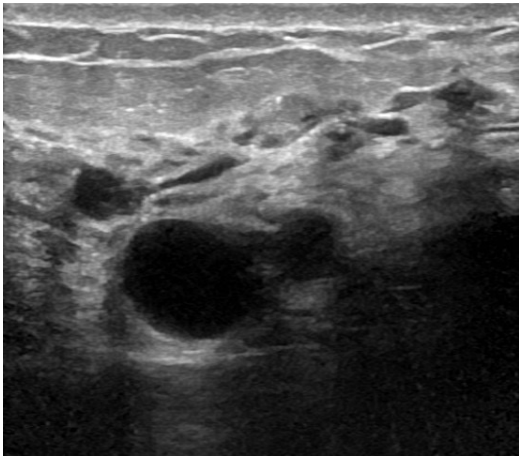
UDIAT (ID: 000117)



UDIAT (ID: 000144)

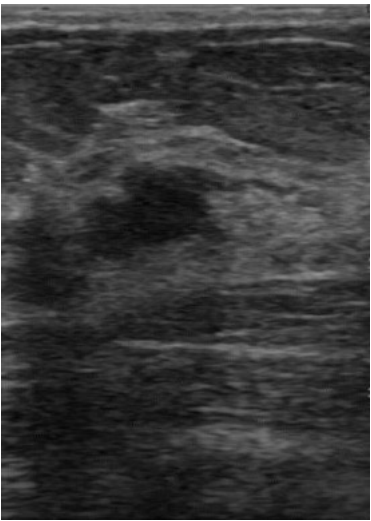


BUSI (ID: malignant (88))



BUSI (ID: benign (41))

| Dataset | #img | Image Resolution |                   | Lesion Resolution |                  |
|---------|------|------------------|-------------------|-------------------|------------------|
|         |      | Min              | Max               | Min               | Max              |
| UDIAT   | 162  | $239 \times 233$ | $791 \times 641$  | $27 \times 18$    | $445 \times 272$ |
| BUSI    | 630  | $190 \times 324$ | $1048 \times 719$ | $37 \times 24$    | $728 \times 444$ |
| BUSBRA  | 811  | $240 \times 295$ | $534 \times 470$  | $28 \times 18$    | $316 \times 381$ |



BUSBRA  
(ID: bus\_0067-r)



BUSBRA  
(ID: bus\_0047-r)

UDIAT: Yap et al., IEEE J. Biomed. Health Inform., 2017.

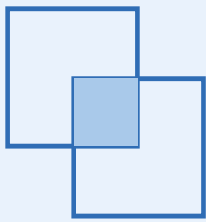
BUSI: Al-Dhabyani et al., Data in Brief, 2020.

BUSBRA: Gómez-Flores et al., Medical Physics, 2024.

# Experimental Settings: Metrics

## Detection metrics

IoU



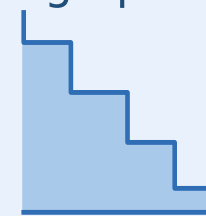
$$|A \cap B| / |A \cup B|$$

F1 score



$$2 \cdot P \cdot R / (P + R)$$

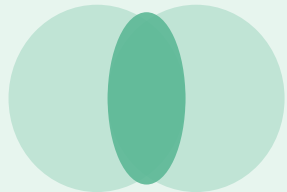
Average precision



area under P-R curve

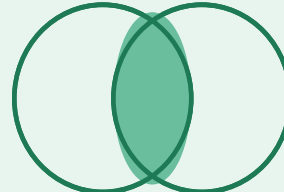
## Segmentation metrics

Jaccard = IoU



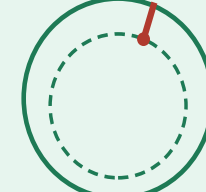
$$|A \cap B| / |A \cup B|$$

Dice



$$2|A \cap B| / (|A| + |B|)$$

HD95



95th-pct boundary dist.



# Experimental Settings: Setup

- **Hardware:**

- Ubuntu 22.04 LTS.
- Intel i9-10940X.
- 64 GB RAM.
- NVIDIA RTX A5000.

- **Software:**

- Python 3.10, PyTorch 2.5.1.
- Ultralytics YOLO version: 8.4.21.
- Segmentation Models PyTorch version: 0.5.0.

- **Training Setup:**

- 80/20 train-validation split.

- **Detection:**

- Architectures: YOLOv3tiny, YOLOv8s, YOLOv12s.
- Trained with bounding boxes derived from lesion masks.

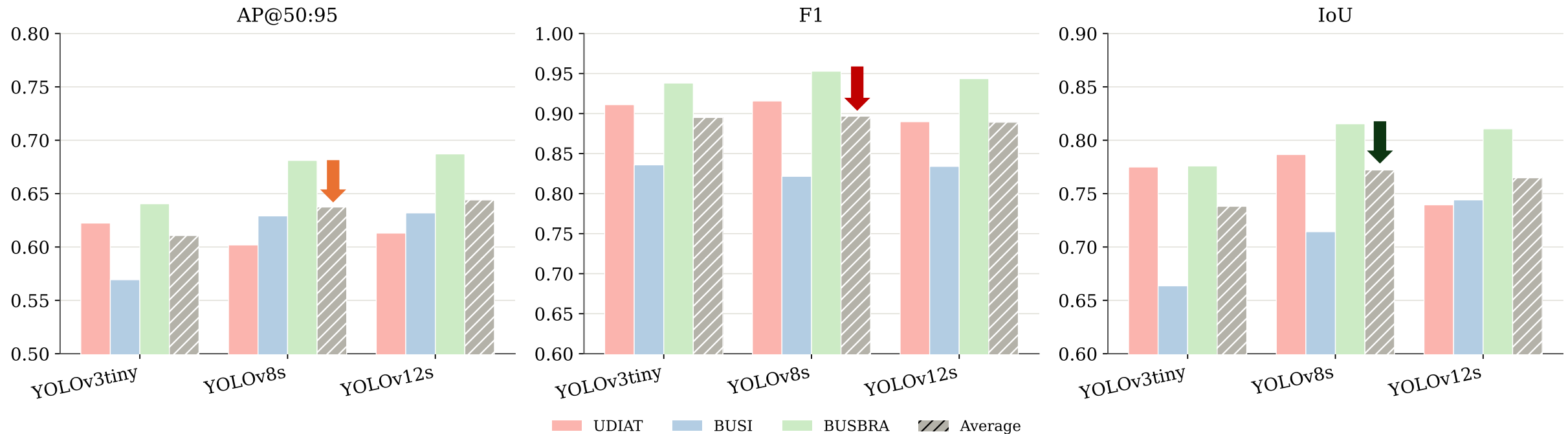
- **Segmentation:**

- Architectures: U-Net, UNet++, and DeepLabV3+.
- Backbone: ResNet-18.
- Optimizer: Adam.
- Learning Rate =  $1 \times 10^{-4}$ .
- Dice loss.



# Results: Quantitative - Stage 1

Detection performance



- **YOLOv8s is the most reliable localizer:**

- ➡• Highest average IoU (0.772).
  - best box-to-lesion overlap.
- ➡• Highest average F1 (0.897).
- ➡• Competitive AP@50:95.

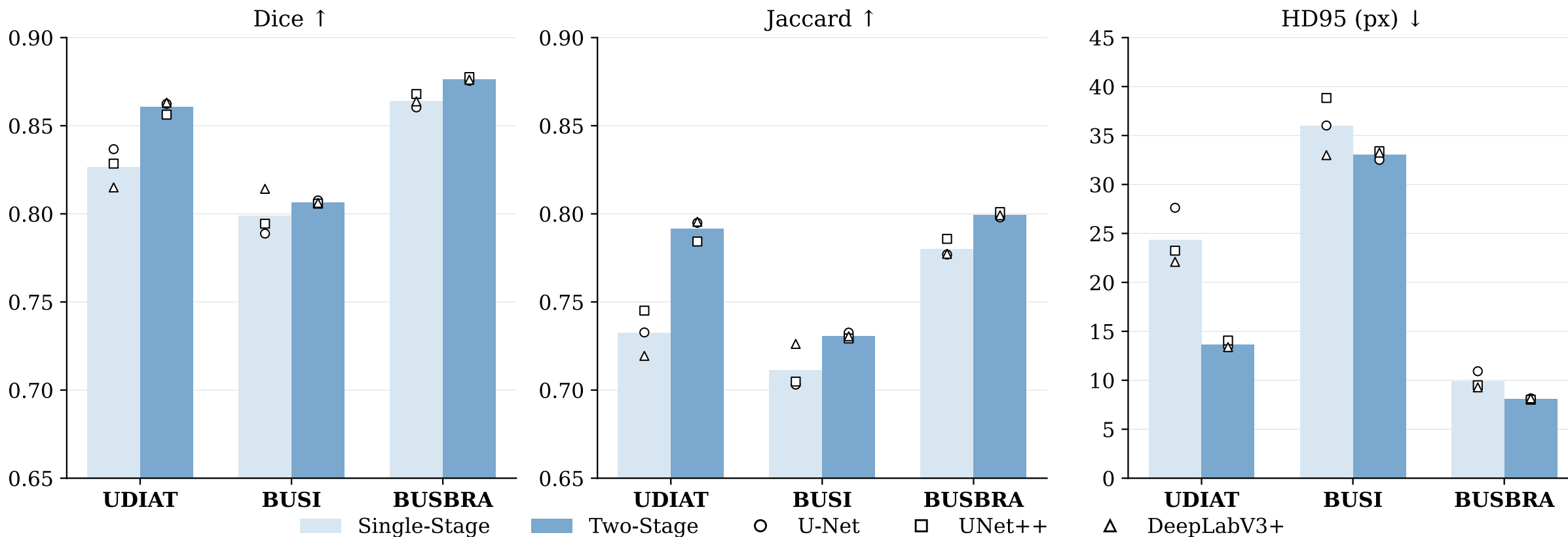
- **YOLOv8s is stays robust across all three datasets:**

- Best on UDIAT and BUSBRA.
- The dependable box to drive coarse-to-fine segmentation.



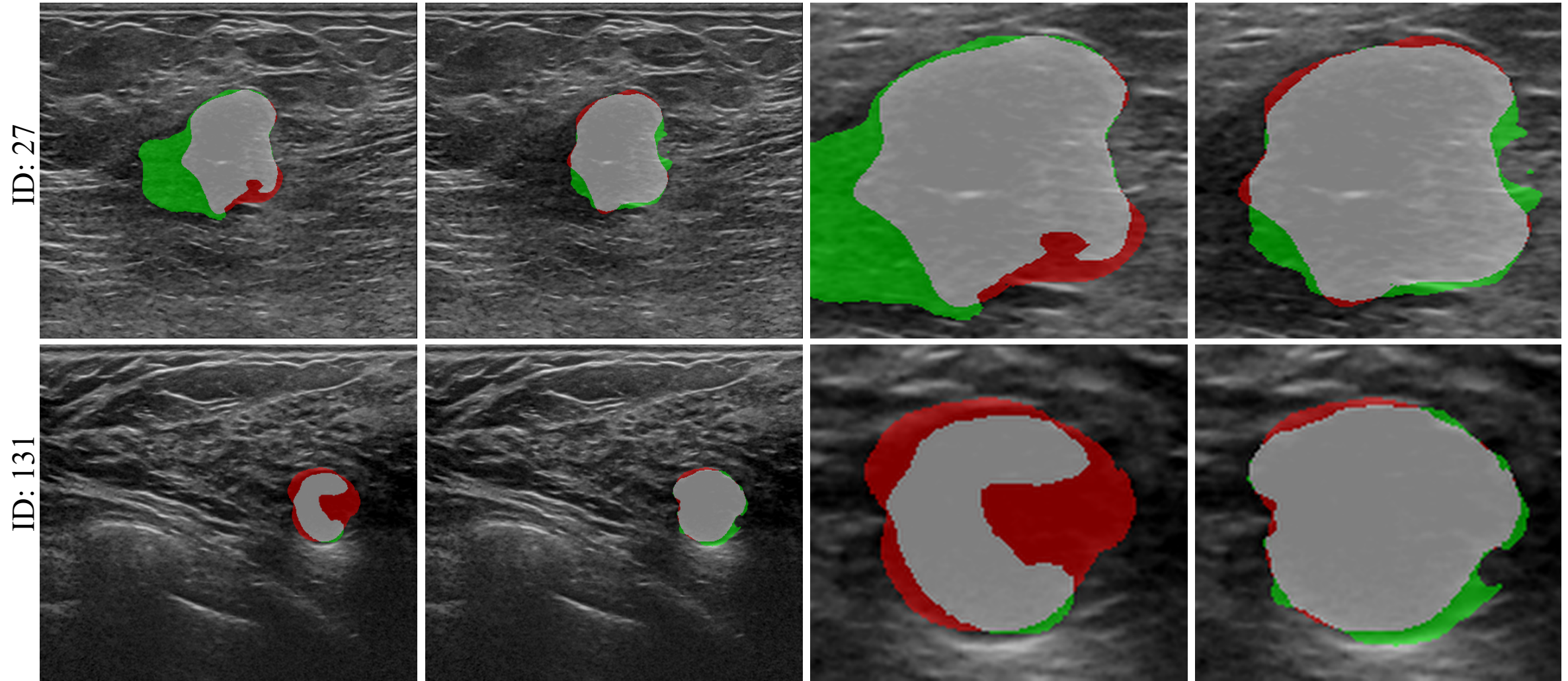
# Results: Quantitative - Stage 1 + Stage 2

Single-Stage vs Two-Stage — average over architectures



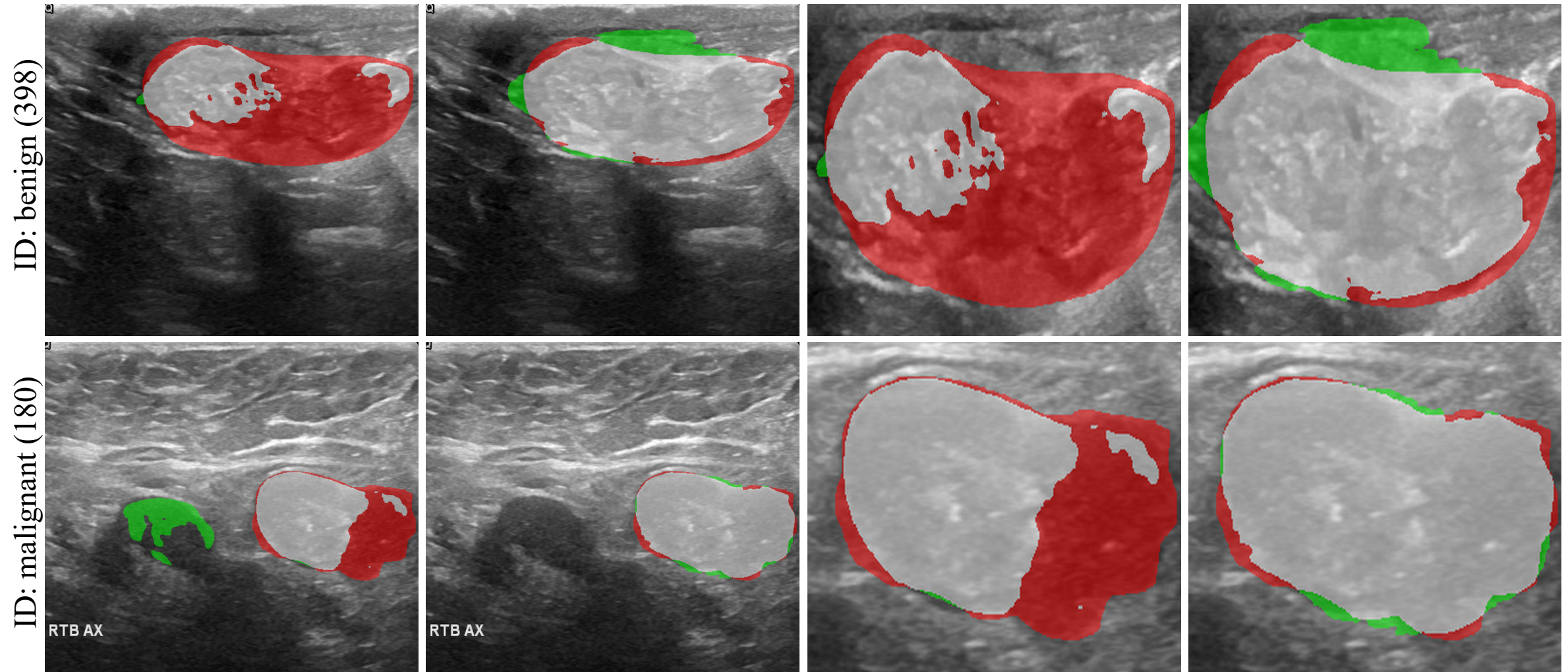
- The two-stage (detection-guided) pipeline outperforms the single-stage baseline on every dataset and metric, with the largest gains on boundary accuracy (HD95).
- Overlaid markers (○ U-Net, □ UNet++, Δ DeepLabV3+) show the improvement holds architecture-by-architecture.

# Results: Qualitative - UDIAT dataset



\* Cols 1–2: single-stage vs. two-stage predictions. \* Cols 3-4: single-stage vs. two-stage zoomed squared size views.  
■ FP: healthy tissue misclassified as lesion. ■ TN: background. ■ FN: missed lesion regions. ■ TP: lesion.

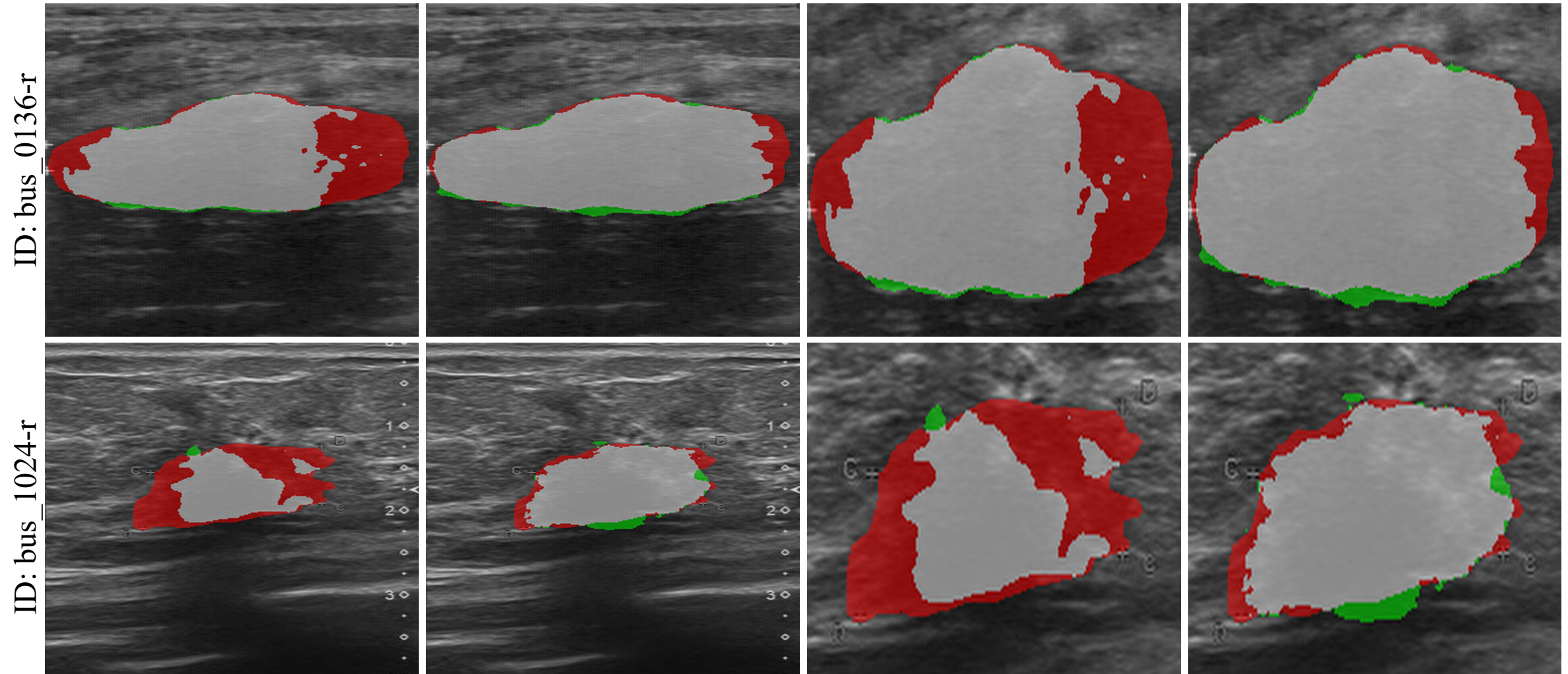
# Results: Qualitative - BUSI dataset



\* Cols 1–2: single-stage vs. two-stage predictions. \* Cols 3-4: single-stage vs. two-stage zoomed squared size views.  
■ FP: healthy tissue misclassified as lesion. ■ TN: background. ■ FN: missed lesion regions. ■ TP: lesion.



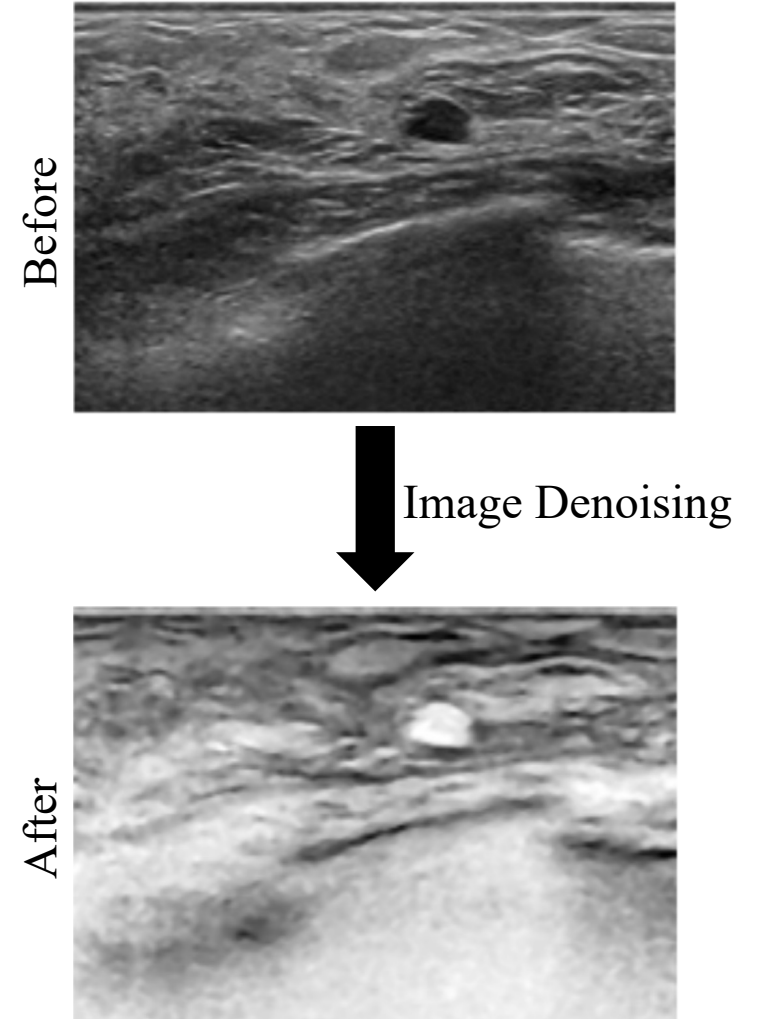
# Results: Qualitative - BUSBRA dataset



\* Cols 1–2: single-stage vs. two-stage predictions. \* Cols 3-4: single-stage vs. two-stage zoomed squared size views.  
■ FP: healthy tissue misclassified as lesion. ■ TN: background. ■ FN: missed lesion regions. ■ TP: lesion.

# Conclusion

- The proposed two-stage framework uses a YOLO model to localise the lesion, then applies a SOTA segmentation model trained on the detected regions.
- Validated across three public datasets, it consistently outperforms conventional single-stage segmentation.
- **Future work:** add a pre-processing enhancement stage to improve image quality before the pipeline.

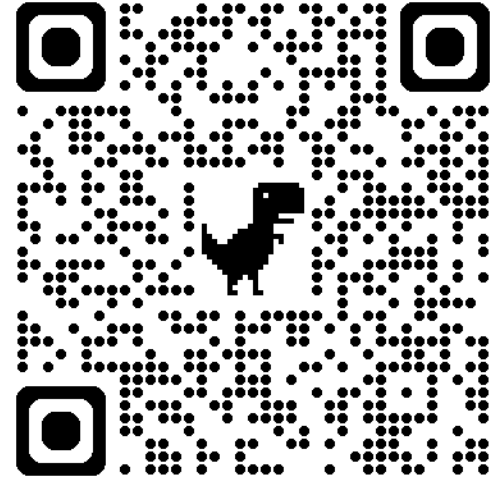


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# Q & A



<https://github.com/mawady/miua26>



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Aamal Alghamdi

[https://www.magnific.com/free-vector/question-mark-symbol-background-with-chat-bubble-design\\_418222512.htm](https://www.magnific.com/free-vector/question-mark-symbol-background-with-chat-bubble-design_418222512.htm)

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